ONE-PAGE BRIEF

Post-quantum cryptography review: halo2 @ 3442dd33754a

Client: Electric Coin Company (Zcash) — public repository (passive review) · Source: github.com/zcash/halo2, commit `3442dd33754a` · Reviewed: 2026-11-06 13:17 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 0 of 4

no post-quantum cryptography

FINDINGS

0 Critical **0** High **2** Medium **0** Low **3** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium — QSI-2026-18047

Quantum-vulnerable Halo 2 (IPA): role not determined (34 file(s)) preliminary

What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...

Medium — QSI-2026-18048

Quantum-vulnerable Sinsemilla: role not determined (9 file(s)) preliminary

What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium role not determined	Halo 2 (IPA), Sinsemilla	Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an...

Static review of source code and manifests; files analysed: 100. No requests were sent to the owner's systems.

Full report, content SHA-256: `3bb0813c2cff59b3362fa0f06a433514f5a3ce8f6feecd0702956a5fd9800cb0`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.